FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY
BACHELOR IN COMPUTER SCIENCE

BCL2113 SOFTWARE ENGINEERING

CONTINUOUS ASSESSMENT 40%

ASSIGNMENT TITLE:
APPLYING SOFTWARE ENGINEERING METHODOLOGIES IN LMS DEVELOPMENT

DUE DATE SUBMISSION:

STUDENT NAME	STUDENT ID

DECLARATION OF INTEGRITY:

I hereby declare that this assessment submission is my own independent work and does not contain plagiarized content, unauthorized assistance, or any form of academic dishonesty. I confirm that I have adhered to the academic integrity policies outlined by the university.

I understand that if any form of academic dishonesty, including plagiarism, falsification of results, or unauthorized collaboration, is detected in this submission, I may face disciplinary actions. This may include, but is not limited to, receiving a failing grade for this assessment or further academic penalties as determined by the university's academic integrity committee.

By signing below, I acknowledge my understanding and agreement with this declaration.

Student 1 Signature: _____ Date: _____
Student 2 Signature: _____ Date: _____
Student 3 Signature: _____ Date: _____

INTENDED LEARNING OUTCOMES

On successful completion of this module, a student will be able to:		Intended Task
1	Explain core principles and practices of software engineering, including software development life cycles, methodologies, and modern engineering tools. (C1, PLO1).	1, 2, 3
2	Apply design patterns and development techniques to create robust and maintainable software systems. (C3, PLO2).	3, 4, 5
3	Evaluate software projects for quality, efficiency, and alignment with user requirements. (C5, PLO3).	6, 7, 8

ASSIGNMENT OVERVIEW

You are part of a software engineering team tasked with designing and developing a Learning Management System (LMS) for a mid-sized private university. The system should allow students to enroll in courses, submit assignments, communicate with lecturers, and access course materials. Lecturers should be able to upload content, manage assignments, track student performance, and communicate with students. The university aims to launch this system in the next academic year.

Propose a system that you feel has many weaknesses and flaws. Write an overview of the system. Support the proposed system by doing good research to include existing issues or problems and solutions to resolve the problem. Include citations and references.

1. Planning and Analysis (10 marks)
   Write an introduction that briefly explain the proposed system. Include 2 objectives.

2. Requirement Engineering (20 marks)
   Conduct requirements elicitation technique(s) from real users to gather data about the system. The questions should be related in getting at least 5 significant requirements. Write a complete User Requirement Specification for the proposed system. This shall include a minimum of 5 functional and 5 non-functional requirements. Apply the requirement engineering processes to ensure requirements are clear, complete, consistent and accurate, free from ambiguity.

3. Software Process Model Justification (10 marks)
   Apply SCRUM methodology for the software development life cycle and describe how it is being implemented during the development process. (No development is needed, just explain the method).

4. System Design UML Diagrams (20 marks)
   Draw a use-case diagram to depicts your functional requirements and class diagram to depicts the objects that is involved for a core LMS function (e.g., assignment submission). Briefly explain your design decisions and components.

5. Team Roles & Project Planning (10 marks)
   Define at least four roles in a software engineering team (e.g., project manager, software tester, developer). Explain how each role contributes to the successful delivery of the LMS. Create a simple Gantt chart or timeline for the development lifecycle.

6. Software Quality and Testing Plan (10 marks)
   Propose a test plan for one key feature (e.g., course enrolment or grading). Include types of testing (e.g., unit, integration, system), sample test cases, and expected outcomes.

7. Risk Management (10 marks)
   Identify at least four potential risks during development. Provide a risk matrix and suggest mitigation strategies.

8. Ethical & Legal Considerations (10 marks)

Discuss any two ethical or legal issues (e.g., data privacy, copyright) relevant to developing an LMS.

SUBMISSION REQUIREMENTS

1. A written report (2000 words) with cover page, table of content, list of figures, list of tables. The report should be well-structured, with clear headings for each section and sub-section.
2. Font type Times New Roman. Font size 12 pts for normal content, 14 pts for heading.
3. Include citation and references using APA or Harvard style.
4. Include diagrams/tables if useful.
5. Report in PDF file. File will be returned if this is not met.

MARKING RUBRIC (100 MARKS)

1. Planning and Analysis (10 marks)

Criteria	Allocated Marks	Descriptors	Marks awarded
Introduction of proposed system is clear and relevant	5	Correct and relevant = 5 marks; unclear = 3–4	
At least 2 objectives stated and well-aligned with system	5	significant and aligned = 5; general but acceptable = 3-4 poor/missing = 0-2	

2. Requirement Engineering (20 marks)

Criteria	Allocated Marks	Descriptors	Marks awarded
Application of suitable elicitation technique(s) with relevant questions	5	**5 marks** – Multiple appropriate techniques applied with highly relevant, well-structured questions. **4 marks** – Appropriate technique applied with mostly relevant questions. **3 marks** – Somewhat suitable technique with a mix of relevant and vague questions. **2 marks** – Technique weakly applied; questions lack focus or depth. **1 mark** – Inappropriate technique with poorly framed or irrelevant questions. **0 marks** – No elicitation technique or questions provided.	
Concluded critical analysis based on data collected	5	**5 marks** – Insightful, logical analysis with strong linkage to user needs and system goals. **4 marks** – Clear analysis with good linkage to user needs. **3 marks** – Some analysis provided but limited depth or weak connections. **2 marks** – Minimal analysis, mostly descriptive without interpretation. **1 mark** – Very weak analysis, little relevance to system requirements. **0 marks** – No analysis presented.	

User Requirement Specification: 5 functional + 5 non-functional	5	**5 marks** – All 5 functional & 5 non-functional requirements are complete, precise, and relevant. **4 marks** – 5 functional + at least 4 non-functional requirements stated, mostly clear. **3 marks** – At least 4 functional + 3 non-functional requirements, some vague or incomplete. **2 marks** – Few requirements provided; lacks clarity or completeness. **1 mark** – Very minimal attempt; requirements poorly stated. **0 marks** – No requirement specification provided.	
Application of requirement engineering processes (clarity, consistency, accuracy, no ambiguity)	5	**5 marks** – Requirements fully demonstrate clarity, consistency, accuracy, and no ambiguity. **4 marks** – Requirements mostly clear and consistent; minor issues present. **3 marks** – Some requirements are clear, but several inconsistencies or ambiguities exist. **2 marks** – Many requirements unclear or inconsistent. **1 mark** – Requirements highly ambiguous and inaccurate. **0 marks** – No application of requirement engineering processes.	

3. Software Process Model (10 marks)

Criteria	Marks	Descriptors	Marks awarded
Explanation of how SCRUM phases (sprint, backlog,	5	3 relevant = 5 marks; 2 or vague = 3–4; <2 = 0–2	

review, etc.) apply to development			
Justification of SCRUM suitability for LMS development.	5	Clear and relevant = 5; partial/generic = 3–4; weak = 0–2	

4. System Design (15 marks)

Criteria	Marks	Descriptors	Marks awarded
Use-case diagram correctly represents functional requirements	5	Clear and complete = 5; mostly accurate = 3–4; incomplete = 0–2	
Class diagram shows relevant objects, attributes, and relationships	5	Correct structure = 5; minor issues = 4–5; major issues = 0–3	
Clear explanation of design decisions and components	5	Clear and relevant = 7–10; general = 4–6; minimal = 0–3	

5. Team Roles & Project Planning (15 marks)

Criteria	Marks	Descriptors	Marks awarded
At least 4 software engineering roles identified	5	Clearly defined = 4-5; vague = 2–3; <4 or unclear = 0–1	
Explanation of role contributions to project success	5	Matched to success = 5-10; limited = 1–4; unclear = 0	
Simple Gantt chart/timeline showing development phases	5	Logical and readable = 4-5; acceptable = 2-3; incorrect = 0–1	

6. Software Quality & Testing (10 marks)

Criteria	Marks	Descriptors	Marks awarded
Selection of one key feature for testing. Clarity and completeness of testing plan	5	Well-defined = 4-5; partial = 2–3; missing = 0–1	
Sample test cases with expected outcomes provided	5	complete = 7-10; partial = 4–6; unclear = 0–3	

6. Risk Management (10 marks)

Criteria	Marks	Descriptors	Marks awarded
Identification of risks	5	Relevant risks = 7-10; general = 3-6 ; < 4 = 0–2	
Mitigation strategies	5	Realistic = 7-10; vague = 3-6; poor/missing = 0-2	

7. Ethical & Legal Considerations (10 marks)

Criteria	Marks	Descriptors	Marks awarded
Identification of issues	5	4 relevant = 7-15; 1 or unclear = 0–6	
Explanation of implications	5	Contextual = 7-15; generic = 4-6; poor/missing = 0-3	

Overall Feedback

Marker:

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Total Mark: _____

Signature	Date